SECTION I
20 Marks
Attempt Questions 1–20
Allow about 35 minutes for this part

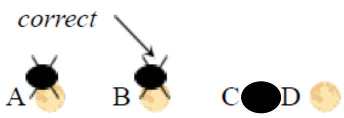
Use the multiple-choice answer sheet for Questions 1–20.

Select the alternative A, B, C or D that best answers the question. Fill in the response circle completely.

Sample 2 + 4 = (A) 2 (B) 6 (C) 8 (D) 9

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:



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1. The Australian Budgerigar is a small Australian parrot which has 26 chromosomes in their skin cells.

   Female birds have a ZW chromosome pair, and male birds have a ZZ chromosome pair. Vets use the bird's DNA to identify its gender.

   Which of the following is a correct statement about Australian Budgerigar birds?

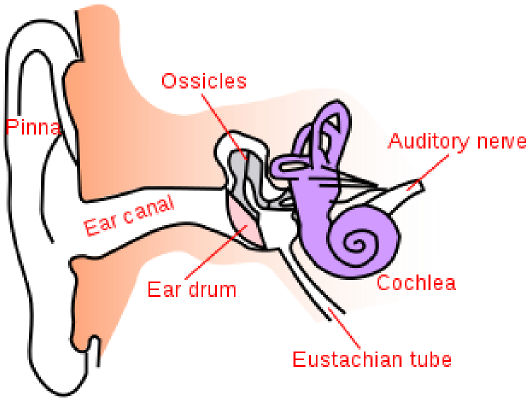
   A. The cells produced by meiosis contain 26 chromosomes.

   B. The cells produced by mitosis contain 26 chromosomes.

   C. The gametes contain 13 pairs of chromosomes.

   D. The somatic cells of female Budgerigar birds contain 13 pairs of autosomes.

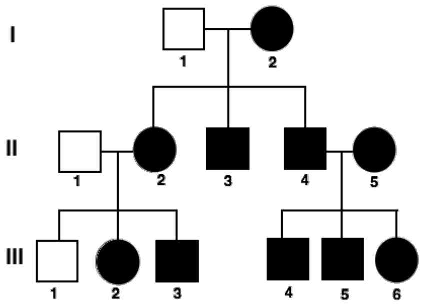
2. The diagram below outlines the structure of the ear:



Which option below correctly matches the structure to its function?

	Structure	**Function**
A.	Ossicles	Coordinate vibrations and then transform the vibrations into electrical impulses.
B.	Eardrum	Long tube that carries sound waves from the outer ear to the middle ear.
C.	Cochlea	Fluid-filled rings that help to determine balance and coordination.
D.	Auditory nerve	Responsible for transmission of electrical impulses to the brain.

3. The pedigree below tracks the presence of dimples through a family's generation. Having dimples is an autosomal dominant trait.



What is the phenotype of the individual **III- 4**?

A. No dimples

B. dd

C. Dimples

D. DD

4. The table below summarises the description of three types of pathogens.

PATHOGEN	DESCRIPTION
A	Range in size from 2 – 1000um. Eukaryotic cell with membrane bound organelles. Reproduce asexually
B	Range in size from 0.5 – 5um. Prokaryotic cell with a single strand of DNA and a cell wall. Reproduce by binary fission
C	Range in size from 30 – 300nm. Acellular containing a protein coat, DNA or RNA. Only reproduce inside a host cell

Pathogen A, B and C respectively are

A. protozoa, bacteria, prion

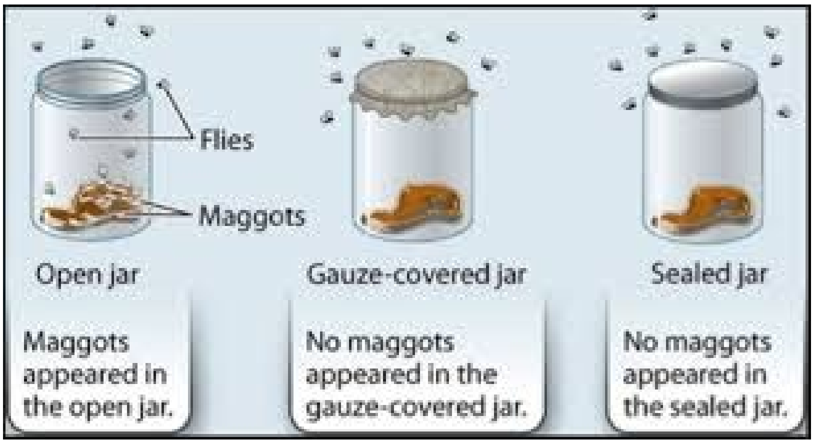
B. bacteria, fungi, virus

C. macroscopic parasite, fungi, virus

D. protozoa, bacteria, virus

Question 5 and 6 refer to the following information.

The following diagram illustrates the experiment conducted in the 1600s by Francesco Redi to investigate the belief that rotting meat spontaneously generated maggots.



5. Which row of the table represents a valid design of the experiment conducted by Francesco Redi?

	Independent Variable	**Dependent Variable**	**Experimental Control**
A.	The amount of flies	The presence of microbes	The open jar
B.	The opening of the jar (open, gauze or sealed)	The presence of flies	The sealed jar
C.	The opening of the jar (open, gauze or sealed)	The presence of microbes	The sealed jar
D.	The size of jar	The presence of microbes	The open jar

6. What is the best explanation for Francesco Redi's maggot experiment?

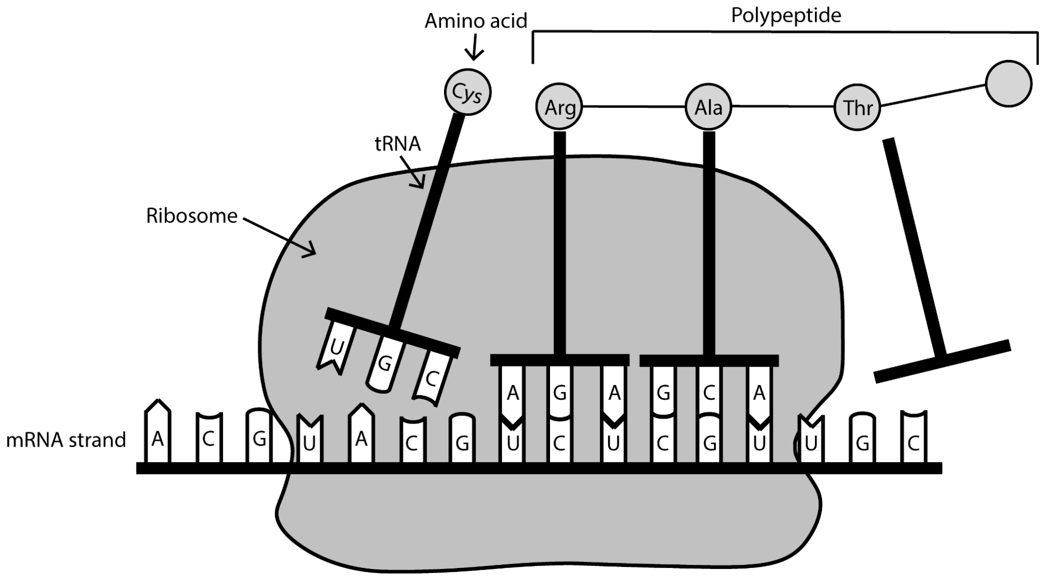
A. Air contains microbes

B. Flies can fit through gauze

C. Cells can appear through spontaneous generation

D. Cells appear from existing cells

7. Below is a diagram of a process during protein synthesis:



Determine the DNA sequence that codes for the amino acid "Ala".

A. C T A
B. C G U
C. G C A
D. G A U

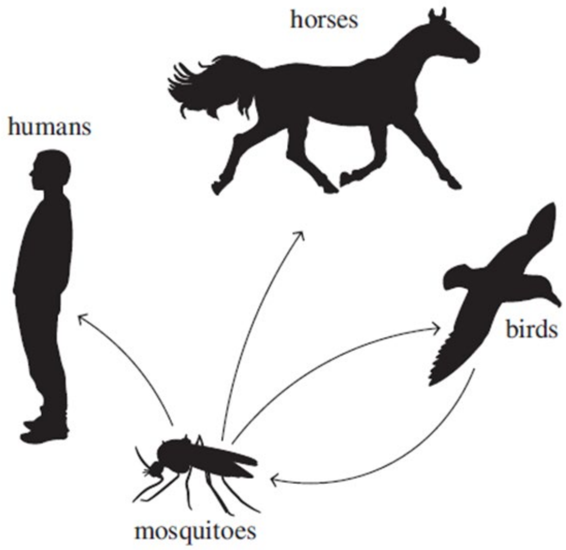
8. Scurvy is a non-infectious nutritional disease caused by a lack of vitamin C in the diet. It causes symptoms such as bruising, bleeding gums, fatigue and severe joint pain.

Davidson and Kakadu plums are sour-tasting, native Australian fruit consumed by Aboriginal and Torres Strait Islander people and as a result, these people had a reduced risk of scurvy.

What can be concluded about the Davidson and Kakadu plums?

A. Davidson and Kakadu plums are likely high in vitamin C.
B. There is insufficient evidence that the plums prevent scurvy.
C. The sourness of the Davidson and Kakadu plum prevents bleeding gums.
D. The Davidson and Kakadu plums are high in calories, providing energy to prevent fatigue

9. The diagram shows the transition of the West Nile virus between various organisms.



Which organism is the vector for the disease?

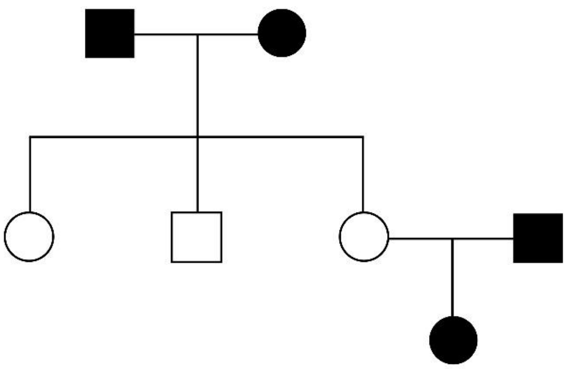
A. birds

B. horses

C. humans

D. mosquitoes

10. The pedigree diagram below shows the inheritance of a condition in a family. The shaded individuals have the condition



Which of the following best describes the inheritance of this condition?

A. Recessive, and not sex-linked

B. Dominant, and not sex-linked

C. Recessive, and sex-linked

D. Co-dominant

11. A scientist has hypothesised that there is an association between people who work with products that contain silica dust (such as sandstone, granite and artificial stone) and a lung disease known as "Silicosis". Silicosis can take anywhere from a few weeks to 10 years to develop.

Symptoms include shortness of breath, dry cough, and tiredness. It also increases the risk of other diseases such as emphysema and lung cancer. It is recommended that people working with these products wear a protective mask and keep the stone wet when cutting to suppress dust.

The scientist tested their hypothesis by following the method below:

1. Randomly select 500 men who work with silica products and 500 men who do not work with silica products.
2. Survey the men every 6 months for 10 years to collect data on diet, occupation, lifestyle and exercise habits. Lung function is also recorded.
3. Record whether or not each man develops silicosis.
4. Statistically analyse the data to identify trends or patterns within the group.

What would be the best way to increase the reliability of this study?

A. Record data from women.
B. Increase the number of participants to 15, 000.
C. Increase the amount of time the study is conducted for 25 years.
D. Record whether each person regularly used the recommended protective measures.

12. Human DNA can be divided into coding and non-coding. Which row of the table outlines the characteristics of coding and non-coding DNA?

	Coding DNA	Non-coding DNA
A.	makes up approximately 98.5% of human DNA	makes up approximately 1.55% of human DNA
B.	codes for proteins	is important to the structure, function and regulation of a cell
C.	can mutate, which often changes the sequence of amino acids	mutations can only be inherited from a parent
D.	can mutate, which often results in phenotypic changes	does not directly code for proteins

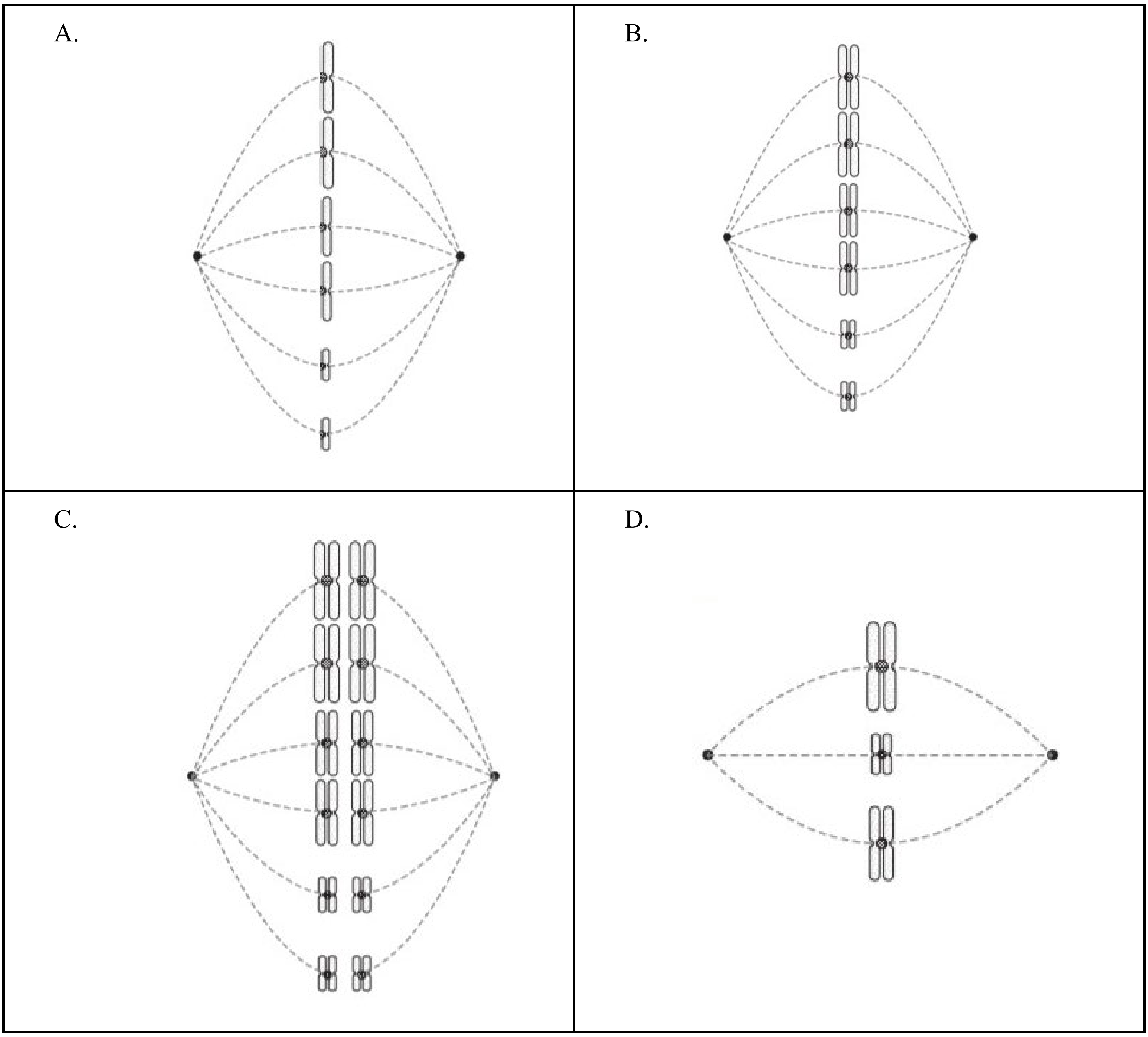
13. During the winter in Antarctica, Emperor penguins clump together by standing within very close proximity of one another. They then slowly move about, rotating the penguins in the centre of the clump to the outside and those on the outside to the centre.

This helps to keep the penguins warm and conserves their energy during a harsh, cold winter.

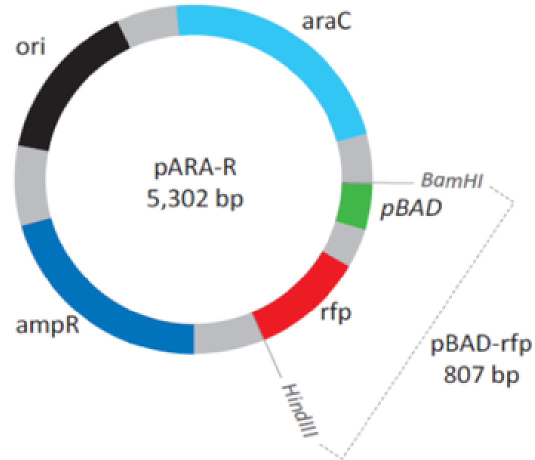
Identify the type of adaptation being demonstrated by the Emperor penguin in this example.

A. Behavioural
B. Physiological
C. Psychological
D. Structural

14. Which diagram correctly models one phase of mitosis that has six chromosomes in its somatic cells?



15. The image below shows the plasmid pARA-R with a desired gene 'rfp'.

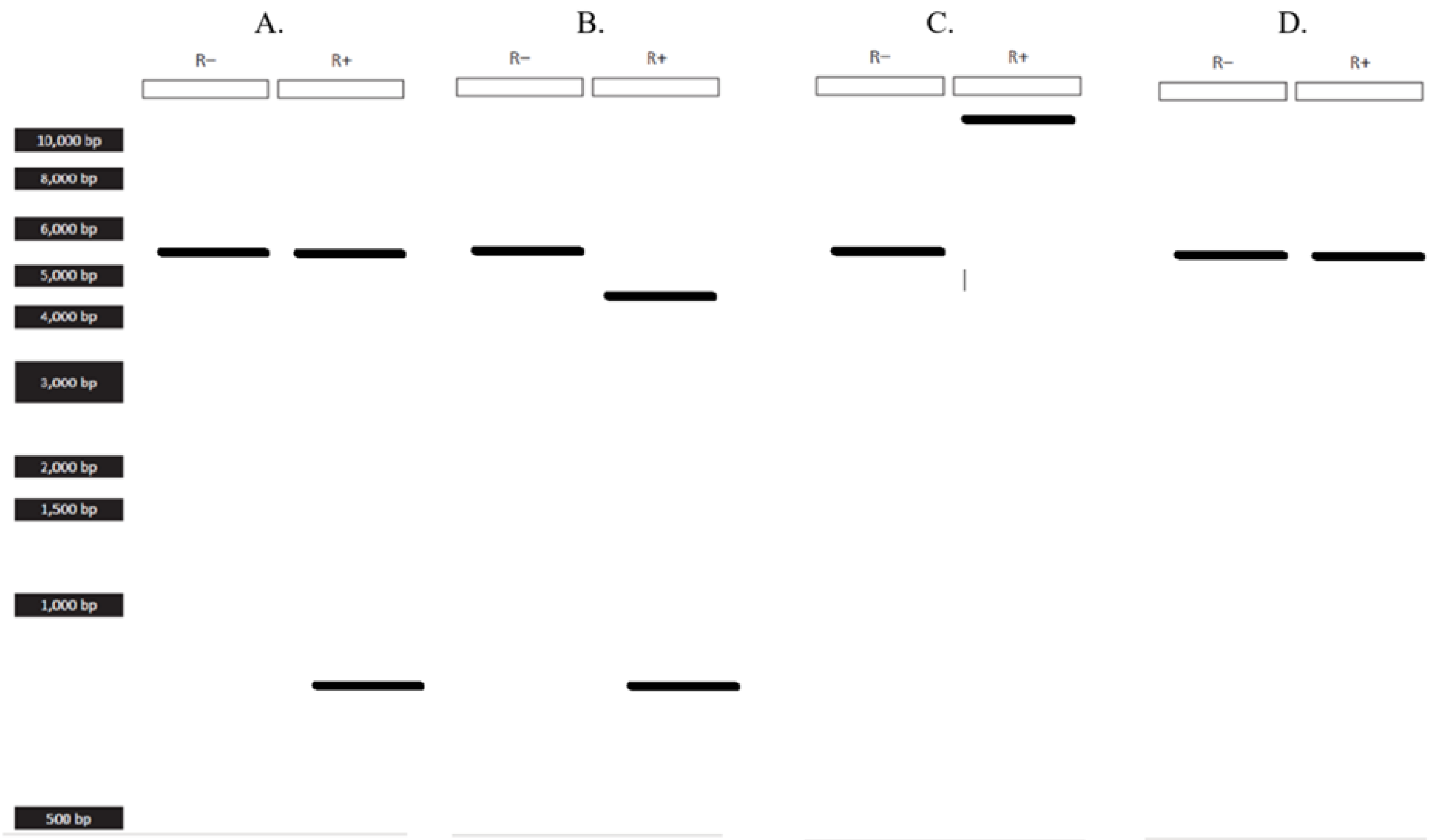


Key	
Ori	Site for initiation of DNA replication
araC	Arabinose activator
ampR	Ampicillin resistance gene
Rfp	Red fluoresce protein gene
pBAD	Promotor
bp	Base pairs

The table below summarises the reagents added to two tubes (R+ and R-). Each tube was centrifuged and left in a 37°Cwater bath for 60 minutes. The results were then analysed.

Reagent	**R+ Tube**	**R- Tube**
pARA-R plasmid	4 µL	4 µL
Restriction enzymes (BamHI & HindIII)	4 µL	
Distilled water		4 µL

Which of the following shows the likely results of the R+ and R- tube?



16. AquAdvantage salmon is a genetically modified organism. The growth hormone- regulating gene in the Atlantic salmon was replaced with the growth hormone-regulating gene from the Pacific Chinook salmon. As a result, the Atlantic salmon can grow continuously year-round, resulting in the ability of the Atlantic salmon to reach market size in 18 months, rather than 3 years.

The steps taken to create the genetically modified Atlantic salmon are listed below:

I	Growth hormone-regulating gene is separated from the plasmid
II	Gene is inserted into a plasmid to be replicated by bacteria
III	Growth hormone regulating gene is isolated from the Pacific Chinook salmon
IV	Cultivated DNA is mixed with fertilised fish eggs and absorbed ensuring the gene is present throughout the fish's body
V	Billions of copies of the growth hormone- regulating gene are produced.
VI	The eggs hatch and some are genetically enhanced whilst others are not. The fish with the new growth hormone regulating gene are used to reproduce.

What is the most likely order of steps that must be taken to produce the genetically modified Atlantic salmon?

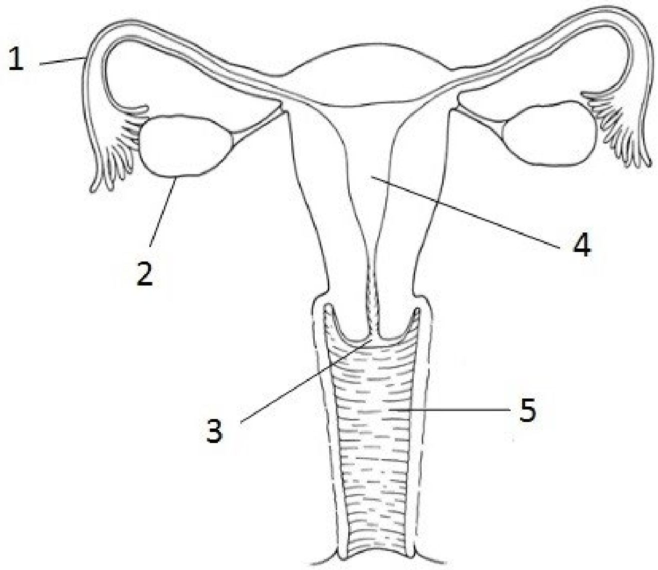
A. III, II, V, IV, VI, I

B. III, V, I, IV, VI, II

C. III, II, V, I, IV, VI

D. III, V, I, II, IV, VI

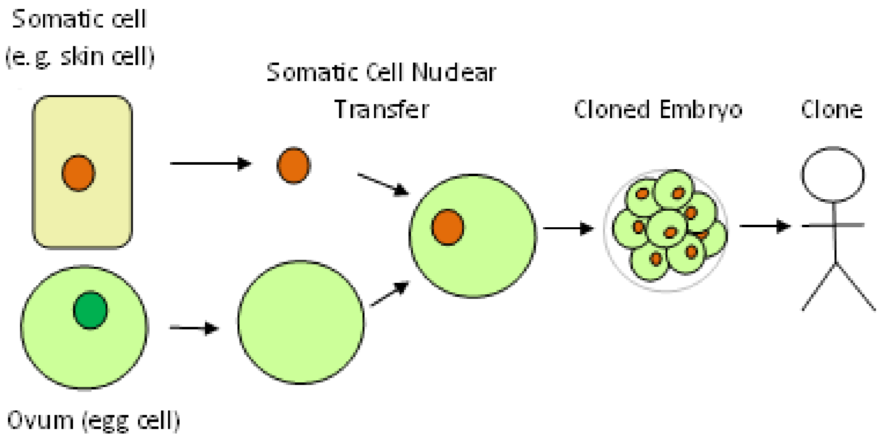
17. The diagram below shows part of the human female reproductive system.



Select the row that correctly shows the labelled diagram to its role in the female reproductive system.

	Site of fertilisation	Site of release of the egg	Site of implantation of the embryo
A.	1	2	3
B.	1	2	4
C.	3	2	1
D.	3	2	4

18. The diagram below shows a process that can be used to clone an organism.



How can you describe the new organism at the end of this process?

A. It will not be viable.

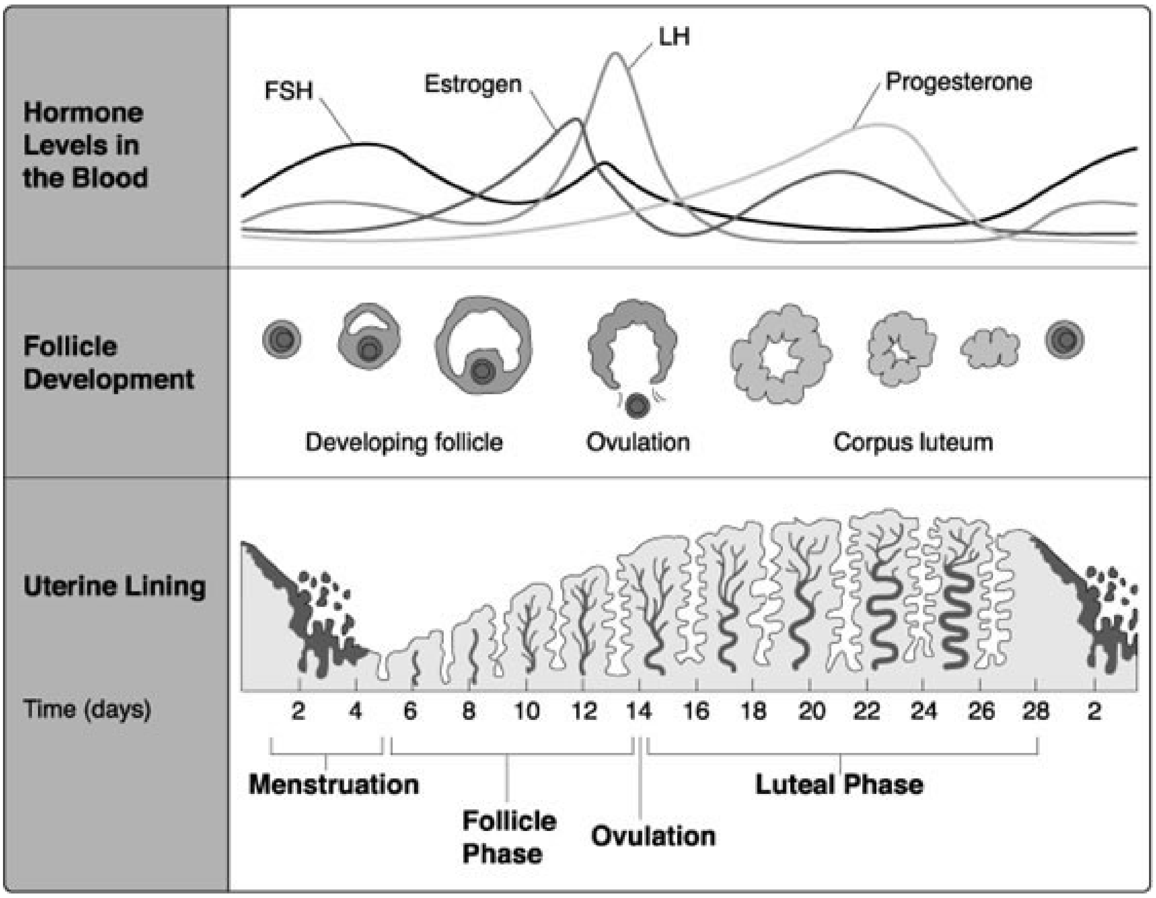
B. It will be genetically identical to the donor somatic cell organism.

C. It will be genetically identical to the donor ovum organism.

D. It will be genetically different from both donors due to variability introduced through sexual reproduction.

Question 19 and 20 refer to the following information.

The diagram below depicts the human menstrual cycle.



19. High levels of oestrogen stimulate a sudden spike in LH and FSH. What major event occurs when LH and FSH spike?

A. Corpus luteum starts to degenerate

B. Menstruation (a period)

C. Uterine lining stop growing

D. Ovulation

20. If the mammalian ovum fails to get fertilised, which one of the following is unlikely?

A. Oestrogen secretion further decreases.

B. Progesterone secretion rapidly declines

C. Corpus luteum will disintegrate.

D. Primary follicle starts developing

End of Section I

Year 12
Trial Examination

YEAR 12 BIOLOGY

SECTION II – LONG RESPONSE
ANSWER BOOKLET

80 Marks
Attempt Questions 21–32
Allow about 2 hours 25 minutes for this part

Instructions	• Answer the questions in the space provided. These spaces provide guidance for the expected length of response • Show all relevant working in questions involving calculations

Question 21 (4 marks)

A female who is heterozygous for a trait reproduces with a male who has the dominant allele for that trait. the likelihood of producing a female with a dominant trait expressed in the phenotype is 100%. The likelihood of producing a male with a dominant trait expressed in the phenotype is 50%.

(a) Identify the pattern of inheritance in this example. (1 mark)

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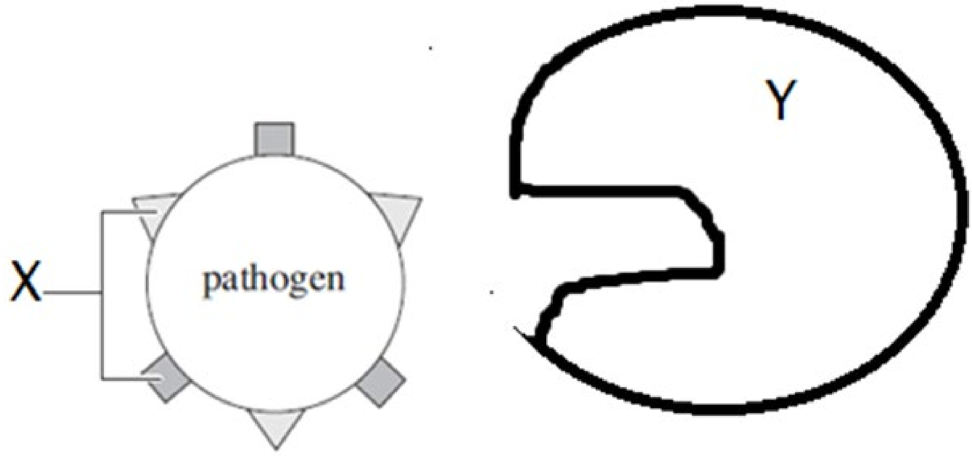
(b) Construct a model to support your answer to part (a). (3 marks)

Question 22 (5 marks)

The diagram below shows some components of the Immune System.



(a) Identify the name of the component labelled X. (1 mark)

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(b) Identify the process represented by the diagram. (1 mark)

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(c) Explain how component X and component Y work together to respond to pathogens. (3 marks)

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Question 23 (8 marks)

Mutations are important factors in genetics.

(a) Identify two types of mutagens. (1 mark)

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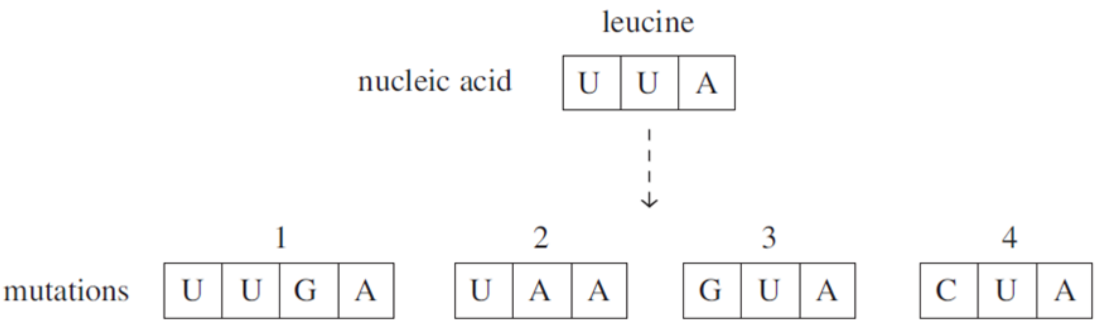
The diagram shows four examples of a type of mutation. The effect of each mutation is shown in the table.



Mutation	*Effect*
1	alters all following amino acids in the sequence
2	acts as a ‘stop’ codon
3	forms valine
4	forms leucine

(b) Identify the type of mutation shown in the diagram. (1 marks)

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(c) Explain why mutation 1 will alter all subsequent amino acids. (2 marks)

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Question 23 continues on the next page

Question 23 (Continued)

(d) Compare somatic mutation and germ-line mutation and their effect on an organism. (4 marks)

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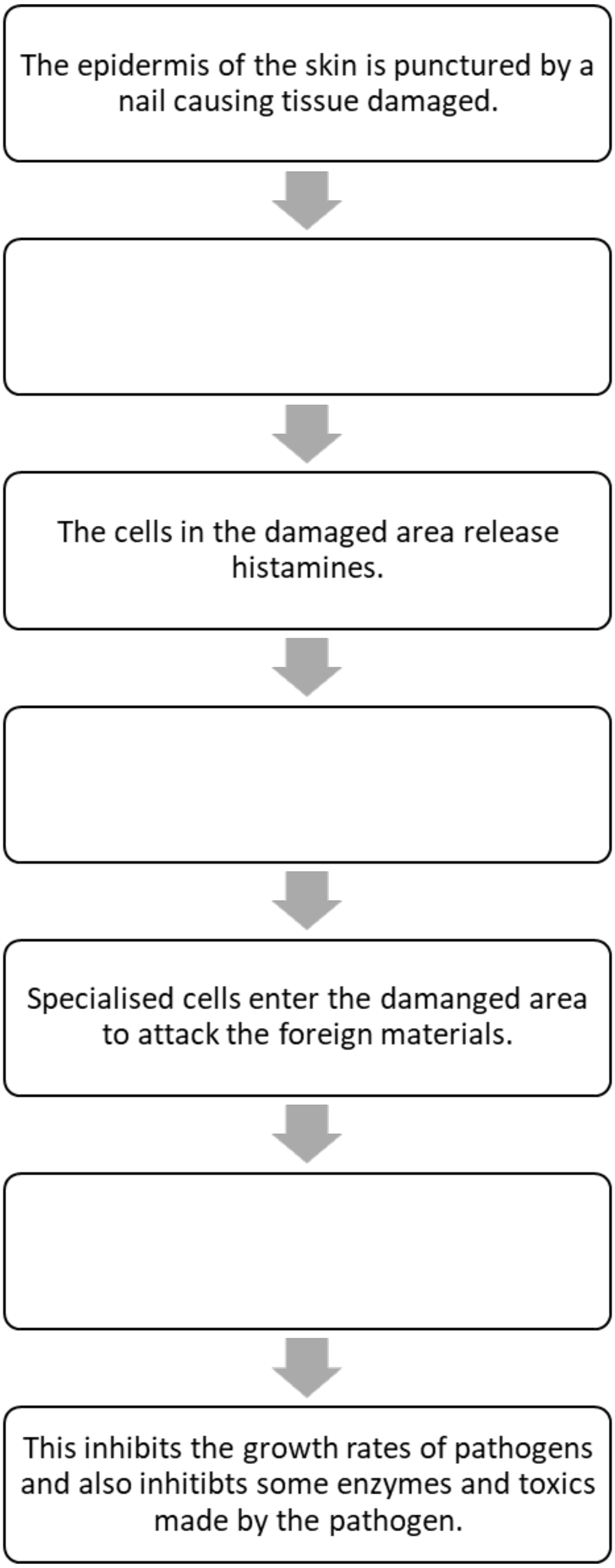
Question 24 (3 marks)

The flowchart below summarises the main steps in the inflammation response. Complete the flowchart.



Question 25 (4 marks)

A disorder occurs when normal tissue is changed, resulting in its structure or function being altered. Technology can assist people who live with disorders. For an individual with a vision disorder, various factors can influence which technology is best suited for the individual.

Technologies used to assist with visual disorders include:

- Spectacles/glasses
- Contact lenses
- Laser surgery

The table below summarises the profiles of two individuals: Jack and Astrid. Both individuals have a visual disorder.

	Individual A	**Individual B**
Name	Jack	Astrid
Age	20 years old	· 68 years old
Employment History	- University student studying a Bachelor of Construction Management	- Retired nurse
Hobbies	· Is a part of the Western Sydney Rugby team	- Visiting cafes with friends on the weekend - Is part of the knitting group in her local community - Reading and playing bingo
Medical History	· Asthma as a child · Hay fever during spring and summer · Has broken a bone in his right arm due to being active as a child	- Type I diabetes - Also has signs of diabetic retinopathy which is caused when high blood sugar damages blood vessels in the retina
Vision Disorder	Short-sightedness, unable to see objects far away. Able to see close up images.	Far-sightedness, objects appear blurry. Able to see objects far away. Also has signs of early cataracts where the lens become cloudy, however she has not experienced any symptoms yet.

Question 25 continues on the next page

Question 25 (Continued)

Using the information from the table, justify which vision disorder technology will best suit each individual. (4 marks).

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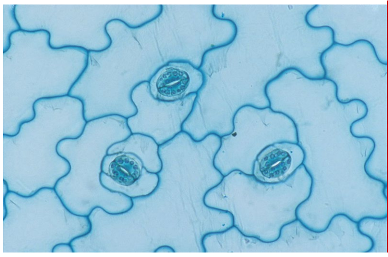
Question 26 (4 marks)

The structures below enable a plant to respond to pathogens.



a) small stomata

b) cell death (apoptosis)

c) thick waxy cuticle

Choose two plant structures to explain how they are disease resistant strategies. (4 marks)

Question 27 (15 marks)

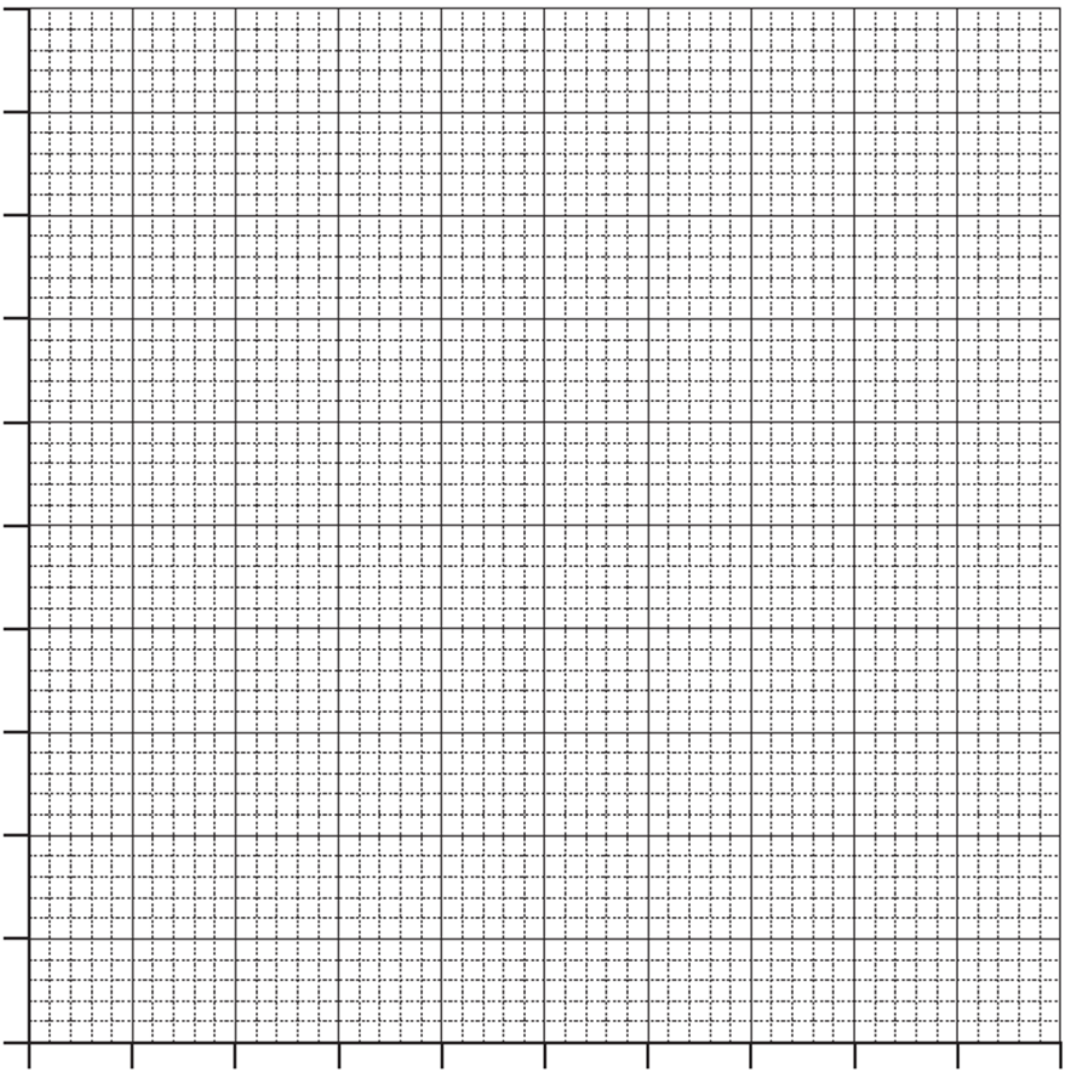
Glaucoma is a disease that damages the optic nerve in the eye. It usually occurs when fluid builds up and increases pressure inside the eye.

A group of researchers conducted a study to investigate a potential link between type 1 diabetes and glaucoma. The researchers selected a sample of participants who have type 1 diabetes and recorded the duration of time that had passed since each participant was diagnosed with type 1 diabetes.

They also recorded the percentage of participants who had experienced glaucoma. The results are shown in the table.

Duration of diabetes (years)	0	5	10	15	20	25	30	35
Percentage of participants with glaucoma	0	2	3	4	6	8	11	13

(a) Draw a suitable graph of the data provided in the table. Include a suitable line of best fit. (3 marks)



- *Question 27 continues on the next page* -

Question 27 (continued)

(b) Construct a hypothesis for the investigation. (1 mark)

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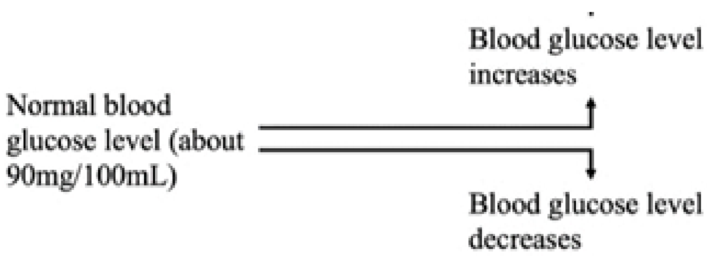
(c) Continue the negative feedback loop shown below to show how glucose levels are regulated when a person is hungry and when they have just eaten a meal. (4 marks)



(d) Type 1 Diabetes is a condition when a person produces little or no insulin. With reference to the feedback loop in (c) explain why a person with Type 1 diabetes must inject insulin when they eat. (2 marks)

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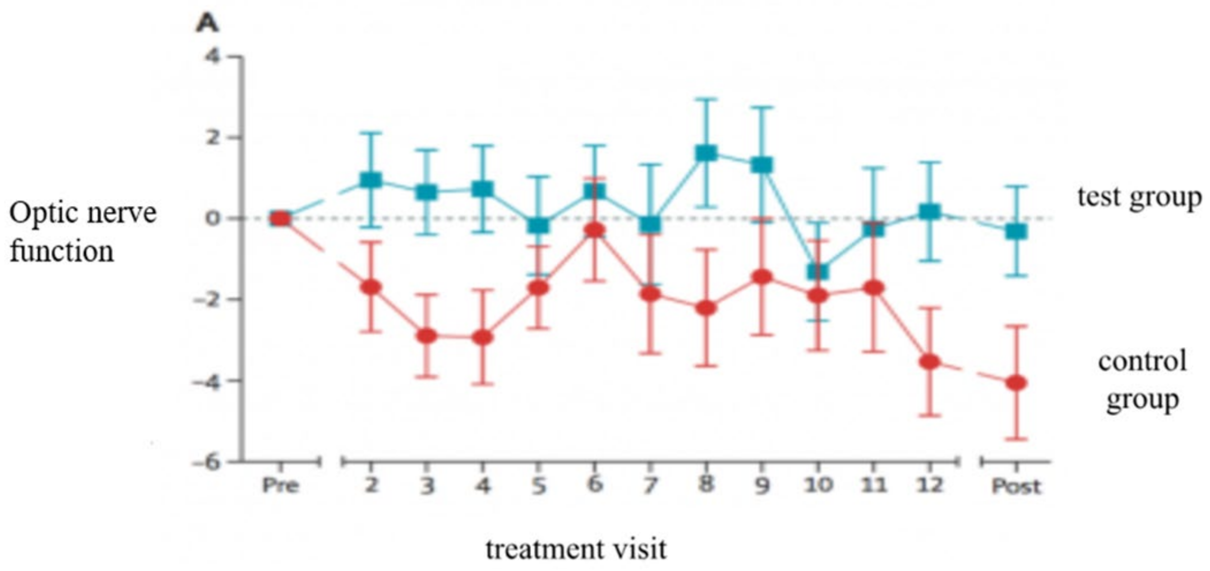
Question 27 continues on the next page

Question 27 (continued)

(e) Doctors have started seeking possible treatment programs for damages to the optic nerve due to glaucoma.

The graph below shows the results of a study on the effectiveness of a glaucoma drug on vision in glaucoma patients. The patients undertaking the study were split into two groups, a test group and a control group. Patients did not know which group they were in. The test group received the treatment drug, and the control group was treated with a placebo (did not contain the treatment drug).

Changes in the average optic nerve of each group are shown in the graph below. Zero on the Y axis represents the optic nerve function (expressed as a %) at the start of the study and the points on the graph shows how the function of the optic nerve then varied from this initial level as the study proceeded. Improved vision is represented by an increase in optic nerve function.



What conclusion can be drawn from the data? Justify your answer. (2 marks)

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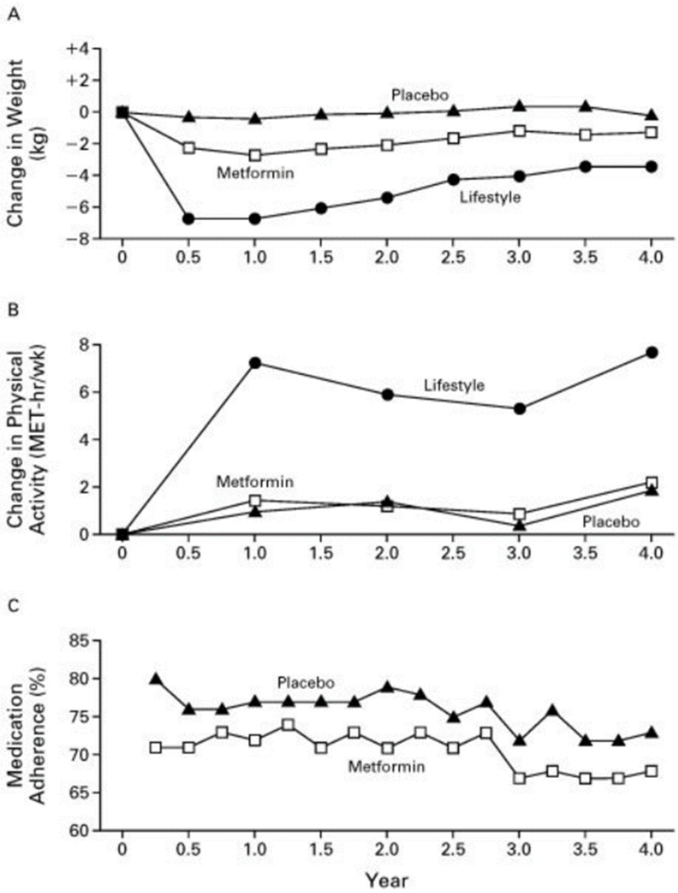
Question 27 continues on the next page

Question 27 (continued)

(f) An epidemiological study on Type 2 diabetes looks at how life-style intervention programs and the use of a drug 'metformin" would prevent or delay the development of Diabetes. 3500 non-diabetic high risk individuals were randomly allocated with one of the following treatment (metformin drug, placebo drug, lifestyle intervention with physical activity).
The graph shows trends in each of the treatment methods over 4 years.



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Note: Medication adherence refers to how well the individual follows the recommended guidelines in taking the medication. A higher medication adherence means that the individual was consistent in following the guidelines on taking the medicine.

To what extent does this data support the use of metformin as a prevention method for Type 2 diabetes?
(3 marks)

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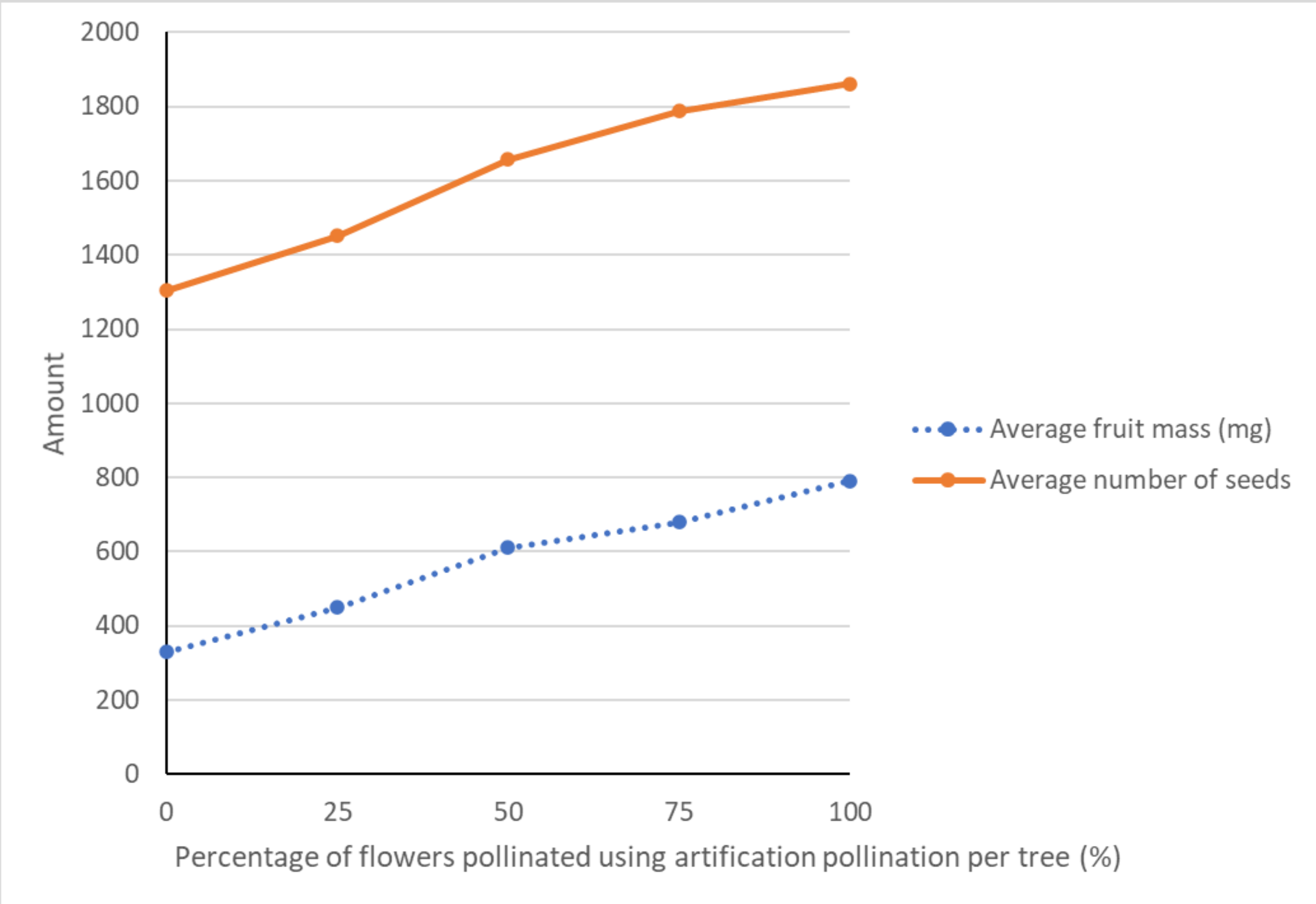
Question 28 (8 marks)

In 2017, some scientists investigated the use of artificial pollination in kiwifruit production. They wanted to find out if artificial pollination (conducted by spraying the pollen directly onto the flowers) of kiwifruits produced larger fruit and more seeds (fertilised ovules) than natural pollination (insect pollination).

The data they collected can be seen in the graph below.



Question 28 continues on the next page

Question 28 (Continued)

(a) Use the data and your own understanding of plant sexual reproduction to evaluate the use of artificial pollination in kiwifruit production. (5 marks)

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Question 28 continues on the next page

Question 28 (Continued)

(b) Many flowering plants can also reproduce asexually as well as sexually.

The three main reproduction method for flowering plants would be:

- Vegetative propagation (asexual reproduction)
- Self-pollination
- Cross- pollination

Explain which reproductive method would you expect to have the greatest frequency of homozygous pairs of alleles? Use genetic crosses to support your answer. (3 marks)

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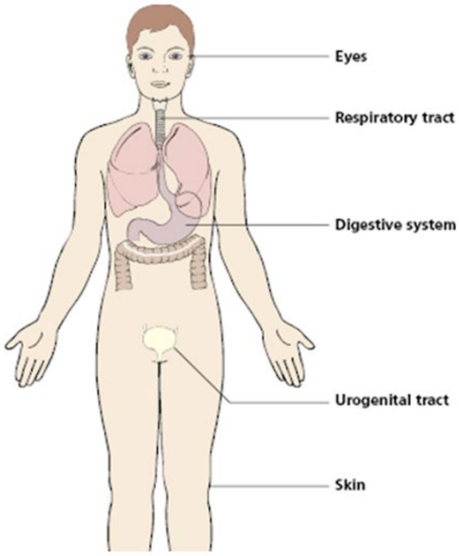
Question 29 (7 marks)
The parts of the human body identified below plays an important part in preventing the entry of pathogens either through physical or chemical means.



a) Using named examples, explain how physical and chemical barriers prevent the pathogens from entering the human body. (4 marks)

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b) Complete the table below for a pathogenic organism which you have studied. (3 marks)

Name of pathogen	
Way in which pathogen is adapted to allow entry into host	
Way in which pathogen is adapted to facilitate transmission.	

Question 30 (7 marks)

A baker has been keeping his freshly baked pastries at a temperature of 30°C for up to one hour before selling them. He wondered whether there would be fewer microorganisms found on the pastries after an hour if he kept them at 65°C. He asked a microbiologist to carry out an experiment to test his idea.

(a) Design a safe and valid method that can be used to test whether the baker's idea was correct. (5 mark)

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Question 30 continues on the next page

Question 30 (continued)

(b) The baker has been experiencing a rash infection on his skin.

The microbiologist suggested using Koch's postulates to identify the cause of the infection.

Describe one limitation in using Koch's postulates for identifying the cause of the infection. (2 marks)

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Question 31 (7 marks)

A student was asked to submit a report evaluating the effect of genetic drift, gene flow and mutations on the gene pool of cheetah population in Africa. The student submitted the response shown.

The cheetah is a vulnerable species of large cat, known for being the fastest land animal. Since the 19th century, it is estimated that the cheetah population has declined from 100,000 to approximately 8,000 cheetahs.

Present-day cheetahs also have low genetic variability, increasing the rate of inbreeding among individuals.

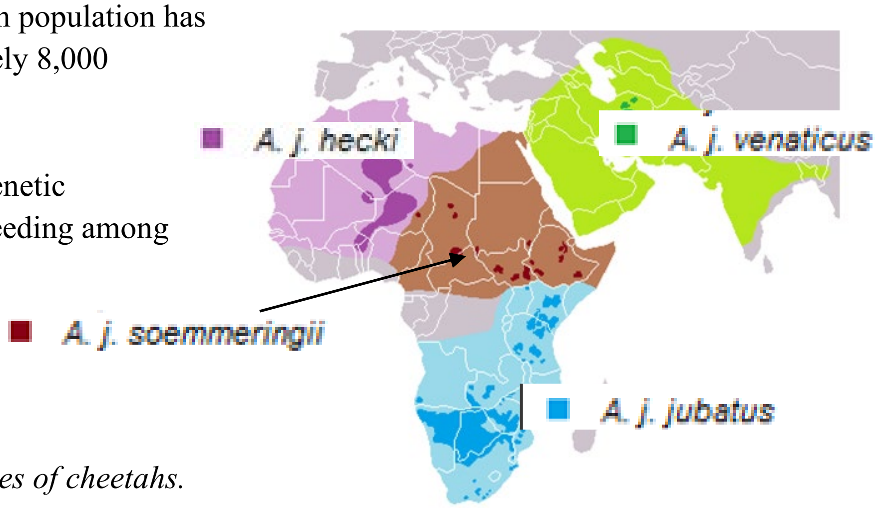


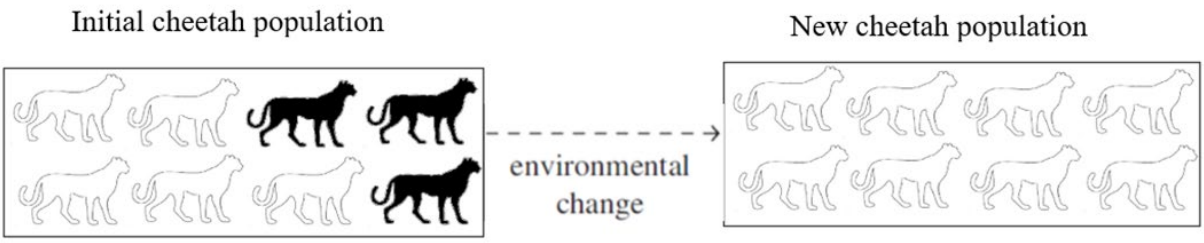
Figure 1: The distribution of subspecies of cheetahs.

From figure 1, we can see that some of the subspecies of cheeta such as the A.J hecki and the A.J soemeringii live close enough for gene flow to occur. We can also hypothesise that A.J hecki is more closely related to A.J venaticus due to the close geographical location than to A.J jubatus.

Genetic drift occurs as a result of random events and is a change in allele frequency within a population (the number of organisms of the same species that live in a particular geographic area at the same time).

This occurs due to a random selection of certain genes, which means the characteristics passed on to an organism's offspring do not necessarily affect their fitness. Genetic drift can also be defined as the transfer of alleles or gametes from one population to another, which lessens the gene pool of the original population. The gene pool is the sum of all the different genes in that population.

The diagram shows how a cheetah population can change over time.



The different phenotypes, and hence genotypes, are represented by the white and black cheetahs. When the cheetah population moves to another environment, some of the cheetahs die (the black ones) and only the white ones are left to breed. This is the reason for all the subgroups of cheetahs as shown in the map in Figure 1.

Genetic drift occurs in both small and large populations. However, its effects are likely to have a greater impact on small populations. This is because the smaller the number of organisms there are in a population, the greater the chances of losing alleles completely. This decreases the size of the gene pool. Mutation are changes in the genetic sequence and produce harmful changes to the cheetahs. This will lead to the cheetahs dying out.

Question 31 continues on the next page

Question 31 (continued)

Using your knowledge of mutation, gene flow and genetic drift to assess the student's report. (7 marks)

Question 32 (8 marks)

Explain how our knowledge of DNA structure and function in prokaryotic and eukaryotic cells have led to development of treatment and prevention for non-infectious diseases. Refer to named examples in your responses. (8 marks)

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End of Section II